

Kuwait University
Department of Mathematics

Calculus I (Math 101)
Second Exam

12 April 2018
Duration 90 minutes

1. [10 pts.] At what point on the curve $y^2 - 1 + 2x = 0$ is the tangent line parallel to the line $y - \frac{1}{3}x = 5$?

Using implicit differentiation, we have $\frac{dy}{dx} = \frac{-1}{y}$. Now the point at which the tangent line is parallel to the given line is the point where $\frac{dy}{dx} = \frac{1}{3}$. Therefore, $y = -3$ and hence $x = -4$.

2. [10 pts.] Find an equation of the normal line to the curve $y = 2\ln(x^2 + x + 1)$ at $x = 0$.

We have $\frac{dy}{dx} = \frac{2(2x+1)}{x^2+x+1} = \frac{4x+2}{x^2+x+1}$. In particular, $\left. \frac{dy}{dx} \right|_{x=0} = 2$, and thus the slope of the normal line is $-\frac{1}{2}$. An equation of the normal line is $y - 0 = -\frac{1}{2}(x - 0)$.

3. [10 pts.] Let $y = (x^2 + 1)^x$. Find $\frac{dy}{dx}$.

We have $\ln y = x \ln(x^2 + 1)$. Now we differentiate both sides with respect to x to obtain $\frac{y'}{y} = \ln(x^2 + 1) + x \left(\frac{2x}{x^2 + 1} \right)$. Therefore, we have $\frac{dy}{dx} = \left(\ln(x^2 + 1) + x \left(\frac{2x}{x^2 + 1} \right) \right) (x^2 + 1)^x$.

4. [10 pts.] Evaluate the following limit; if it exists.

$$\lim_{x \rightarrow 0} \frac{\sin(3x) + \tan(2x)}{6x} = \lim_{x \rightarrow 0} \left(\frac{\sin(3x)}{6x} + \frac{\tan(2x)}{6x} \right) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

5. [10 pts.] The radius of a circle is decreasing at a rate of 2 cm/min. At what rate is the area of the circle changing when the area is 25π cm²?

The area of a circle is: $A = \pi r^2$. Therefore, $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

When $A = 25\pi$ cm², $r = 5$ cm. Hence, $\left. \frac{dA}{dt} \right|_{r=5} = (2\pi)(5)(-2) = -20\pi$ cm²/min.

6. [10 pts.] Let $f(x) = \frac{x-1}{x^2-3}$

(a) Find the linearization of f at $x = 1$.

(b) Use part (a) to estimate $f(1.02)$.

We have $f'(x) = \frac{(x^2-3) - (2x)(x-1)}{(x^2-3)^2} = \frac{-x^2+2x-3}{(x^2-3)^2}$. So $f'(1) = \frac{-1}{2}$.

Therefore, $L(x) = 0 + \frac{-1}{2}(x-1) = -\frac{1}{2}(x-1)$.

Now $f(1.02) \approx -\frac{1}{2}(1.02-1) = -0.01$.

7. [10 pts.] Find the absolute maximum and absolute minimum values of $f(x) = xe^{-x}$ on $[0, 2]$.

We have $f'(x) = e^{-x}(1-x)$, so f has one critical number $x = 1$. Now $f(0) = 0$, $f(1) = \frac{1}{e}$ and $f(2) = \frac{2}{e^2}$.

Thus, f has absolute maximum value $\frac{1}{e}$ and absolute minimum value 0.

8. [10 pts.] Verify that $f(x) = 2x^2 + 3x + 4$ satisfies the hypotheses of Rolle's Theorem on the interval $\left[-2, \frac{1}{2}\right]$.

Then find all numbers c that satisfy the conclusion of Rolle's Theorem.

It is clear that f is continuous on $\left[-2, \frac{1}{2}\right]$ and differentiable on $\left(-2, \frac{1}{2}\right)$ (f is a polynomial). Also, $\boxed{f(-2) = 6 = f(1/2)}$.

Therefore, by Rolles's Theorem there exists a $c \in \left(-2, \frac{1}{2}\right)$ s.t. $\boxed{f'(c) = 4c + 3 = 0}$. Thus, $\boxed{c = \frac{-3}{4} \in \left(-2, \frac{1}{2}\right)}$.

9. [$4 \times 5 = 20$ pts.] In each of the following multiple choice questions, select the correct answer.

I. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{2x} =$

a) $\frac{1}{2}$.

b) $\boxed{\frac{1}{\pi}}$.

c) 1.

d) None of the above.

II. If $h(x) = g(f(x))$, where $f(2) = 3$, $f'(2) = 4$, $g(3) = 6$, and $g'(3) = -5$, then $h'(2) =$

a) 12.

b) 24.

c) $\boxed{-20}$.

d) None of the above.

III. If $\sin^{-1} x + e^y = e$, then $\left. \frac{dy}{dx} \right|_{(0,1)} =$

a) 1.

b) 2.

c) $\boxed{\frac{-1}{e}}$.

d) None of the above.

IV. $\lim_{x \rightarrow 0} \frac{\cosh x - 1}{x} =$

a) $\boxed{0}$.

b) 1.

c) e .

d) None of the above.
